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Effective e-commerce price prediction with machine learning technologies

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ABSTRACT

The development of e-commerce plays a significant role in improving market efficiency and promoting the optimization of industrial structures. Machine learning technology has been widely used in the exploration and prediction of data in this field. In this study, we aim to investigate the performance of machine learning models for commodity price prediction problems. We used eight different machine learning models to predict commodity prices and discounted prices, and added Bayesian optimization and sentiment analysis to improve the performance of the model. Experimental results showed that RandomForest model had the best comprehensive and prediction performance. In addition, we also found that Bayesian optimization could effectively improve the performance of the models, and the sentiment analysis method did not bring a positive effect in the experiment of predicting discounted prices.

CCS CONCEPTS

• **General and reference** → Cross-computing tools and techniques; Estimation.

KEYWORDS

E-Commerce, Machine Learning, Regression, Sentiment Analysis

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1 INTRODUCTION

With the rapid development of the Internet, e-commerce has been fully integrated into all areas of production and life. The development of e-commerce plays a significant role in improving market efficiency and promoting the optimization of industrial structures. According to data from the Department of E-commerce and Information Technology of the Ministry of Commerce, global e-commerce sales are expected to exceed US$5 trillion for the first time in 2022, accounting for 1/5 of the overall retail sales. In 2022, the number of online shopping users in China will reach 845 million, accounting for 79.2% of the total Internet users. Online retail sales of physical goods were 11.96 trillion RMB, accounting for 27.2% of the total retail sales of consumer goods. It can be seen that e-commerce and online shopping have become an important part of the consumer market and an indispensable way of life and consumption for people.

The rapid development of e-commerce has brought great challenges to both management and information systems, and the traditional manual management method can no longer meet growing business development. Moreover, the labor cost of hiring more people is increasing.

Therefore, it is of great significance to use machine-learning methods to explore and predict data in the e-commerce field. Compared to traditional manual analysis methods, machine learning methods play a more crucial role and contribute to fields such as personalized services, supply and demand forecasts, and price trend forecasts through automated, high-efficiency, personalized, and large-scale data analysis methods.

In this study, we aim to investigate the performance of machine learning models for commodity price prediction problems. We used eight different machine learning models to predict commodity prices and added Bayesian optimization and sentiment analysis to improve the performance of the model. The experiment was conducted in two steps. First, we explore the dataset of mobile phone sales on JD.com and predict the price of mobile phones by analyzing the characteristics of mobile phones. Then, by analyzing a product sales dataset on Amazon, we extracted the data features before discounts and predicted the price after discounts. Finally, the overall performance of the model in the two experiments was comprehensively evaluated to determine the machine learning models with stronger generalization ability and better prediction performance in predicting price problems. Predicting the price of goods and the price after discounts can help companies optimize pricing strategies, formulate effective promotional activities, increase sales revenue, and meet consumer needs through more accurate market positioning, thereby enhancing competitiveness.

2 RELATED WORK

2.1 Machine learning in electronic commerce

Machine learning has the power to process data and extract critical information from large-scale data to help e-commerce businesses make informed decisions. E-commerce data often has complex patterns and associations, and by using machine learning algorithms, we can better discover the patterns and regularity hidden behind the data to grasp market trends. Currently, e-commerce industries are becoming increasingly competitive. Machine learning models, which are automated and efficient in data processing, enable

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companies to make predictions and judgments faster and adapt to a changing market. These models exhibit good performance and satisfactory prediction results.

Pavlyshenko [1] constructed a stacked approach to single-model regression ensembles that combined the results of different predictive model analyses with different feature sets into one result. The results show that the stacking method can improve the accuracy of the validation and out-of-sample sets and the performance of the forecast model in sales time series forecasting.

Iftikhar et al. [2] proposed a fusion model of machine learning and business intelligence using three machine learning algorithms to predict product prices: multi-class random forest, multi-class logistic regression, and multi-class one-to-many. Finally, it was found that multi-class one-to-many had a higher accuracy in both prediction preservation and cross-validation.

Kopczewska et al. [3] used a transactional dataset from Olist e-commerce retail in Brazil to build and test a comprehensive customer churn model by exploring typical behavioral data (including purchase value, shipping cost, product category, etc.), and adding an analysis of the customer's location and socioeconomic environment. The basic model used was XGBoost and Logistic regression, and it was found that the propensity to churn depends on the value paid for the first order, the number, type, and transportation cost of the items purchased, as well as the demographic environment and location. The results of this study have important developmental implications for CRM (customer management system) development.

Cao et al. [4] constructed a financial early warning model based on deep learning algorithms, and used financial indicators as inputs to predict risks. The experimental results showed that the accuracy of the model in predicting whether a company is in crisis can be maintained at more than 73%, and the more years of data used, the higher the accuracy.

Pan et al. [5] proposed a method to predict sales volume by automatically extracting effective features from structured time-series data using convolutional neural networks. The product attribute information and related logs are used as inputs, and then the features are automatically extracted by the method. Finally, these features are input into the last layer of the convolutional neural network to make predictions. Compared with traditional machine learning methods, this method avoids time-consuming and labor-intensive human intervention.

Zhou [6] proposed an e-commerce product advertisement recommendation method based on deep learning and a distributed representation. First, a similarity network was constructed based on the distribution of advertisement topics, and then a deep learning model was constructed to predict the click rate of advertisements. Subsequently, a time window is introduced to control the data transfer in the hidden layer of the neural network to improve the traditional recurrent neural network. The experimental results show that the proposed model is more accurate than the traditional model.

Satu [7] collected a dataset of online shoppers' willingness to buy from a public data warehouse, performed multiple feature root transformation methods, and used different state-of-the-art classifiers (Decision Tree, Random Forest, etc.) on the dataset to complete the modelling of online customers' willingness to buy behavior. Finally, the different results of the proposed model were analyzed. The experimental results show that Random Forest has the highest accuracy compared to other models and is the best method for identifying the customer purchase rate.

Salamzadeh et al. [8] collected relevant data from Iranian and Hungarian users on online social platforms through questionnaires, and used a Gaussian mixture model (GMM) and multi-layer perceptron (MLP) machine learning algorithms as methods for consumer clustering and classification to explore the popularity of grocery applications. The experimental results showed that Wolt in Hungarian and Snappfood in Iran are the most popular programs.

Xiao et al. [9] proposed a model using a recurrent neural network (RNN) algorithm to predict the influence trend of sales activities based on key information between e-commerce platform group users and marketing activities. Experiments show that the model can predict user behavior in the context of sales activities and dynamically depict changes in activity influence trends.

However, the number of models used in these studies was small. In this study, we used eight models, boosting, bagging, and regression algorithms, to comprehensively evaluate the results.

2.2 Sentiment analysis in electronic commerce

Sentiment analysis is the process of extracting and understanding emotions, attitudes, and feelings from text, voice, or images using technologies such as natural language processing, text mining, and machine learning. User feedback is an important source of information in the e-commerce world. Through sentiment analysis, we can obtain feedback such as users' experience and satisfaction, which is conducive to user experience optimization and product improvement, so that enterprises can better understand the market reaction and adjust marketing strategies in time. Currently, sentiment analysis methods are widely used in e-Commerce and work well.

Chen et al. [10] collected data from nearly 200,000 sales records of a cross-border e-commerce company, and built an artificial intelligence framework. This framework combines text mining, WaveNet neural networks, multiple regression, and SHAP models to explain the impact of product descriptions on sales forecasts. Experiments show that this model has a higher forecasting performance than traditional time-series forecasting methods and forecasting models that do not consider product descriptions.

Hakami et al. [11] collected user review data from e-commerce websites in Saudi Arabia, and proposed a new deep learning model using dataset analysis and a neural computation dual-attention model (DL-DA). The DL-DA model uses lexical analysis and deep-learning methods. Lexical analysis detects lexical features in customer reviews that emphasize sustainability. They performed a deep learning analysis by obtaining approximately 323,150 customer reviews from Arabic e-commerce websites as an experimental dataset. The experimental results show that this model can effectively improve the accuracy of lexical text analysis. The accuracy of the proposed model is 96.5%.

In many cases, the literature above believes that the sentiment analysis is helpful for prediction, but our results show that for the prediction of discounted prices, user review sentiment scores do not help or even have a counterproductive effect on improving model prediction performance.

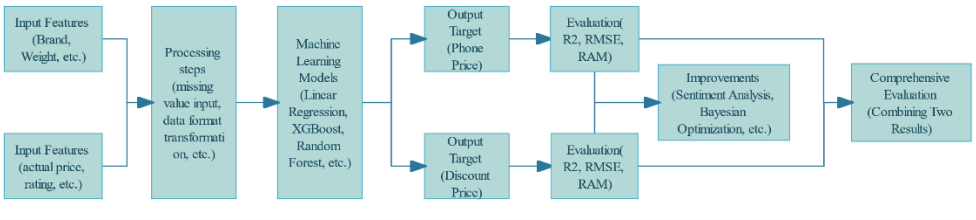

Figure 1: The workflow of this study.

3 PROBLEM DESCRIPTION

3.1 Overview

This study focuses on the price prediction problem. The target of this prediction problem is mobile phone prices and discounted prices, which are of great interest to customers, especially those who are price sensitive. In the first part of the question, we can predict the final price of the mobile phone by inputting its different characteristics. In the second part, we use the relevant characteristics before the discount to predict the discount prices. Through the prediction and exploration of the original and discounted prices, we conducted research on price prediction problems. As show in Figure 1.

3.2 Phone dataset description

In this part of the research, an open-source dataset from Kaggle was used, which contained mobile phone sales data of 17 brands on JD.com. The dataset has 1199 rows (i.e., different sales records) and 21 columns (i.e., features). In this article, we use six characteristics (i.e., battery, brand, price, rear camera, resolution, screen size, and weight) of mobile phones and the prices of the mobile phones for analysis.

Figure 2 shows a heat map representing the strength of the relationship between certain features. From the figure, we can see that the strongest correlations with price are RAM and ROM. The correlation with comments was very weak.

3.3 Discount price dataset description

In this part of the research, an open-source dataset from Kaggle, which contains over 1,000 Amazon products in India, was used. The dataset had a tabular format with 1465 rows (i.e., different products) and 16 columns (i.e., features). In this article, we use the product price, product rating, number of reviews, discounted price of each piece of sales data in the data set as research objects.

Figure 3 shows a paired graph representing pairwise relationships between discounted_price and actual_price. From the figure, we can see that most of the data points are concentrated in the lower price area, indicating that the prices and discounted prices of the items in this data set are mainly concentrated in the low-price range. In addition, after being classified into three areas of 'low', 'middle' and 'high' according to the initial price, there is a certain linear relationship between the two features in all three areas.

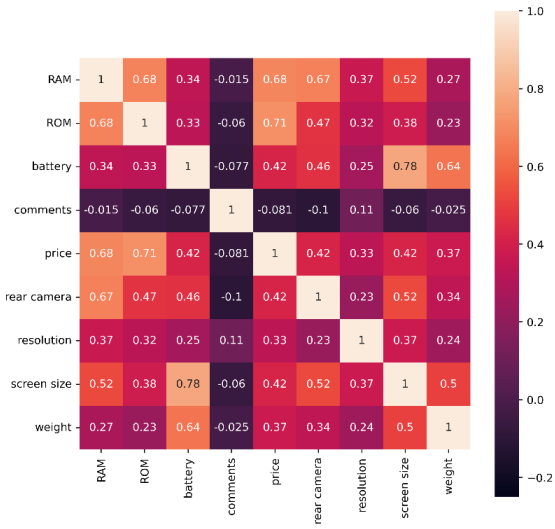

Figure 2: Heatmap showing the correlation among features.

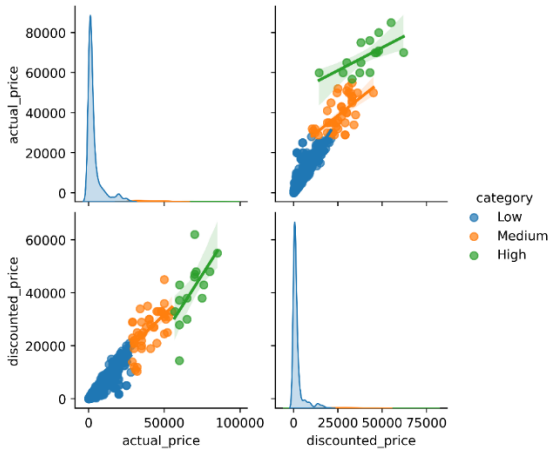

Figure 3: Pairwise relationships between discounted_price and actual_price.

3.4 Data preprocessing

On the Phone Dataset, we use the Re module to parse natural language for some information so that useful numerical information

can be extracted. Then, through the statistics of missing values in all columns, we found some columns with many missing values, such as CPU freq and CPU cores. Therefore, we extracted several relatively complete feature columns from the original dataset that are closely related to price, and finally determined 519 rows of valid sales data as the research object. Each row of data has seven features (price, rear camera, weight, battery, resolution, screen size, and brand). To facilitate the input of features, we also performed one-hot encoding on the "brand" column to convert this categorical variable into multiple feature columns in binary form for use in machine learning models. This ensures that the model understands and utilizes brand information without mistakenly assuming a sequential relationship between brands.

In the stage of processing the discount price dataset, the data type of some features (i.e., actual_price, discounted_price, rating_count) are altered to be used for correct data analysis. Then, a strange product rating is modified to 4.0 according to the product page through product_link. By splitting the list of category features, two new columns as the main category and sub-category are created and added. Likewise, the strings in the main category and sub-category were transformed accordingly. By splitting the strings in the columns of user_id and user_name, these two columns are displayed as one id per row.

Finally, we deleted an abnormal data with a large price to prevent it from interfering with the experiment.

In addition, the standard scaler method is also applied in preprocessing to achieve standardization, so that machine learning models are able to better fit and learn from these two datasets.

4 MODEL DESCRIPTION

4.1 Sentiment analysis method description

At this stage, the NLTK package was called in to provide a library for processing natural language data. A sentiment analysis language package called Vader was then downloaded from the NLTK library. The SentimentIntensityAnalyzer class in the sentiment module is a pretrained model that can be used to analyze the sentiment of a given text. It uses a dictionary-based approach to calculate the sentiment score for a given text. Scores were calculated based on the presence of certain words in the text and their association with positive or negative sentiments.

4.2 Machine learning models

We used eight models for prediction: XBoost, Catboost, Randomforest, Linear Regression, Lasso Regression, Ridge Regression, AdaboostRegressor, and LightGBM.

Within the realm of regression, we leveraged Linear Regression as a foundational model, assuming a linear relationship between input features and the target variable. Additionally, we employed Lasso Regression, which introduces L1 regularization, fostering sparsity in the model and aiding in feature selection. Ridge Regression, another variant used, incorporates L2 regularization to handle multicollinearity and enhance model stability.

In the boosting category, both XGBoost and AdaboostRegressor were applied. XGBoost, known for its extreme gradient boosting capabilities, sequentially builds decision trees, each correcting errors from the previous one, leading to a robust predictive model. AdaboostRegressor, on the other hand, iteratively constructs a series of weak learners, giving more weight to misclassified instances in each iteration, thereby improving the overall model performance.

Random Forest, a model falling under the bagging category, was employed for its ensemble learning approach. By training multiple decision trees on bootstrap samples of the data, Random Forest achieves enhanced predictive accuracy through aggregation.

Lastly, LightGBM, a gradient boosting framework, was utilized for its efficiency in tree learning. Its leaf-wise growth strategy and histogram-based learning contribute to faster training and improved performance, especially beneficial for large-scale datasets.

5 EXPERIMENTS

5.1 Settings

In experiments on the phone dataset, owing to the small amount of data, it is not suitable to directly divide the dataset. To prevent overfitting, we used cross-validation on the dataset in our study. The dataset into 5 equal sized. Each time, four subsets were used as the training set and the remaining set was used as the test set. This is repeated five times, and each subset becomes a test set. Finally, the average performance of the five evaluations is calculated.

In experiments on the discounted price dataset, the dataset is randomly divided into a training set and a test set according to a ratio of 70% to 30% in common machine learning practice.

In addition, the R2, RMSE, and MAE indicators were used to evaluate the performance of the models.

Because of the variability in scores for the RandomForestRegressor, AdaBoostRegressor, and Catboost models for each run, we conducted multiple runs for each model, calculated the average scores from these runs, and organized these average scores into a table.

5.2 Results

5.2.1 Phone dataset results. Input: weight, rear camera, screen size, brand, battery, resolution.

From Table 1, we can see that the performance of RandomForest, LightGBM, CatBoost, and XGBoost is better than other models.

5.2.2 Discounted price results. Input: actual_price and rating and rating_count.

From Table 2, we can see that the four models (Linear Regression, Lasso Regression, Ridge Regression and XGBoost) perform the best.

5.2.3 Overall evaluation. We combined the model's predicted performance on the two datasets and averaged the scores.

It can be seen from Table 3 that the Random Forest model performs the best in terms of overall performance, with an R2 index of 0.8574. The three models, XGBoost, CatBoost, and LightGBM, also showed good overall performance.

5.2.4 Improvements suggestions. For the problem of mobile phone price prediction, we used Bayesian optimization to improve the performance of the model. Bayesian optimization is an iterative optimization method for global optimization. Its main idea is to improve the estimation accuracy of the objective function by continuously selecting new points in the search space for evaluation,

Table 1: The results of the phone dataset.

Model	R2	RMSE	MAE
Linear regression	-8.4607	3862975106623407.50	3862975106623407.5
Lasso regression	0.3953	1177.82	791.20
Ridge regression	0.3931	1180.31	793.25
XGBoost	0.7164	838.84	347.19
CatBoost	0.7511	795.27	336.95
LightGBM	0.7529	794.63	374.87
AdaBoostRegressior	0.4529	1127.31	815.52
RandomForestRegressior	0.7913	737.15	326.82

Table 2: The results of discounted price dataset.

Model	R2	RMSE	MAE
Linear Regression	0.9239	2152.00	903.40
Lasso Regression	0.9239	2153.03	903.39
Ridge Regression	0.9239	2153.21	903.44
XGBoost	0.9340	2004.27	766.32
CatBoost	0.9194	2215.52	818.06
LightGBM	0.9056	2397.86	861.90
AdaBoostRegressor	0.9018	2446.13	1395.70
RandomForestRegressor	0.9236	2156.84	796.94

Table 3: The overall evaluation results.

Model	R2	RMSE	MAE
Linear Regression	-3.7684	1.93E+15	1.93E+15
Lasso Regression	0.6596	1665.425	847.295
Ridge Regression	0.6585	1666.76	848.345
XGBoost	0.8252	1421.555	556.755
CatBoost	0.8352	1505.395	577.505
LightGBM	0.8292	1596.245	618.385
AdaBoostRegressor	0.6773	1786.72	1105.61
RandomForest Regressor	0.8574	1446.995	561.88

so as to find the global situation more effectively. We used Bayesian optimization to find better hyperparameter combinations for each model (except linear regression) and brought them back to the models for experiments.

From Table 4, we can see that the performance of all the models improved accordingly after hyperparameter optimization.

For the problem of discount price prediction. We use the Sentiment Intensity Analyzer model to analyze user reviews for each piece of data and return a composite sentiment score called a compound. The compound score corresponds to the sum of the valence scores of each word in the lexicon, providing a single measure of polarity. Its value ranges between -1 (indicating the most extreme negative sentiment) and +1 (indicating the most extreme positive sentiment).

From Table 5, we can see that the sentiment analysis method did not bring a performance improvement to models.

Table 4: The results with Bayesian optimization.

Model	R2	RMSE	MAE
Lasso Regression	0.4408	1148.26	791.39
	(+0.0455)	(-29.56)	(+0.19)
Ridge Regression	0.4422	1146.73	796.44
	(+0.0491)	(-33.58)	(3.19)
XGBoost	0.8128	701.49	332.36
	(+0.0964)	(-137.35)	(-14.83)
CatBoost	0.7944	734.05	369.13
	(+0.0433)	(-61.22)	(+32.18)
LightGBM	0.8103	706.03	337.06
	(+0.0574)	(-88.60)	(-37.81)
AdaBoostRegressor	0.5305	1040.40	693.25
	(+0.0776)	(-86.91)	(-122.27)
RandomForest	0.8235	687.29	325.62
Regressor	(+0.0322)	(-49.86)	(-1.20)

Table 5: Added compound results.

Model	R2	RMSE	MAE
Linear Regression	0.9240	2151.81	901.70
	(+1E-04)	(-0.19)	(-1.70)
Lasso Regression	0.9240	2151.85	901.71
	(+1E-04)	(-1.18)	(-1.68)
Ridge Regression	0.9240	2152.02	901.75
	(+1E-04)	(-1.19)	(-1.69)
XGBoost	0.9174	2242.85	804.86
	(-0.0166)	(+238.58)	(+38.54)
CatBoost	0.8780	2726.04	933.17
	(-0.0414)	(+510.52)	(+115.11)
LightGBM	0.8989	2481.39	881.03
	(-0.0067)	(+83.53)	(+19.13)
AdaBoostRegressor	0.8973	2500.55	1550.99
	(-0.0045)	(+54.42)	(+155.29)
RandomForest	0.9188	2223.58	836.91
Regressor	(-0.0048)	(+66.74)	(+39.97)

6 CONCLUSION

This study used eight models, namely XBGoost, CatBoost, Adaboost-Regressor, RandomForest, LightGBM, Linear regression, ridge regression, and lasso regression, to predict the price of mobile phones and the discounted prices of other products, respectively. The random forest model performed well in predicting mobile phone prices. The XGBoost model exhibited the best performance in predicting the discounted price. Combining the two prediction results, we conclude that the RandomForest model has the best comprehensive and prediction performance among the models. Additionally, we attempted to improve the performance of the model. In the experiment to predict mobile phone prices, we added Bayesian optimization to determine better hyperparameter combinations for each model. The experimental results showed that after adding Bayesian optimization, the prediction performance of the eight models correspondingly improved. We also added sentiment analysis to the study on predicting discounted prices. The features of sentiment scores were added by analyzing comments in the sales data. The experimental results show that adding sentiment analysis does not help model prediction, but instead reduces the prediction performance. Thus, we can indicate that on this dataset, sentiment analysis does not have a significant effect on predicting discounted prices.

Our experiments also had some shortcomings. Predictions of price and discounted price used two datasets, respectively, which makes the entire study not completely continuous and lacks overall convincing. Additionally, due to limited open data, we did not obtain more data on consumer reactions after discounts, such as sales volume. If the data is more comprehensive and richer, we can expect to do an in-depth analysis of what happens after the discount. Subsequently, we were able to model the sales markdown strategy and develop strategies to improve the profit level.

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